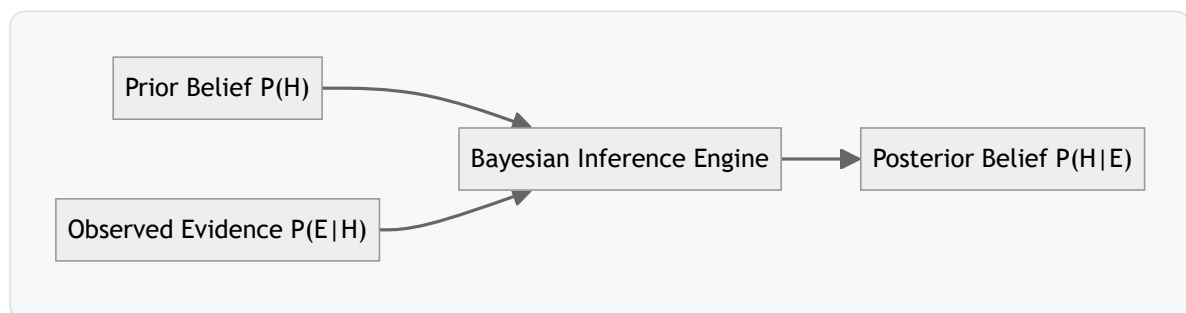


Day 20 - Lecture 20.12: Bayes' Theorem

[Python 3.10+](#)[PDF notes.pdf](#)[Language: English](#) | [हिंदी / Hinglish](#) | [Back to Lecture 20 Module](#)

1. Overview & Epistemic Belief Updating

Bayes' Theorem is the foundational engine of modern Bayesian machine learning, probabilistic inference, and cognitive artificial intelligence. It provides the exact mathematical framework for updating a degree of belief in a hypothesis (H) after observing empirical evidence (E).



2. Mathematical Derivation & Formulation

From the definition of conditional probability:

$$P(H \cap E) = P(H | E) \cdot P(E)$$

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Equating the right-hand sides yields **Bayes' Theorem**:

$$P(H | E) = \frac{P(E | H) \cdot P(H)}{P(E)}$$

Decomposition into Scientific Components:

1. **Prior Probability** ($P(H)$): Degree of belief in hypothesis H *before* observing evidence E .
2. **Likelihood** ($P(E | H)$): Probability of observing evidence E assuming hypothesis H is true.
3. **Marginal Evidence** ($P(E)$): Total probability of observing evidence E across all possible hypotheses, expanded via the Law of Total Probability:

$$P(E) = P(E | H)P(H) + P(E | H')P(H')$$

4. **Posterior Probability** ($P(H | E)$): Updated degree of belief in hypothesis H *after* incorporating evidence E .

Full Expanded Formulation:

$$P(H_i | E) = \frac{P(E | H_i)P(H_i)}{\sum_{j=1}^k P(E | H_j)P(H_j)}$$

3. Real-World Demonstrative Example

Continuing our factory example from Lecture 20.11:

- Factory 1 (B_1): produces 50% ($P = 0.50$), defect rate 1%
- Factory 2 (B_2): produces 30% ($P = 0.30$), defect rate 2%
- Factory 3 (B_3): produces 20% ($P = 0.20$), defect rate 5%
- Total Defect Rate: $P(D) = 0.021$

A customer receives a **defective** speaker (D). What is the posterior probability that it originated from **Factory 3** (B_3)?

$$\boxed{P(B_3 | D) = \frac{P(D | B_3) P(B_3)}{P(D)} = \frac{0.05 \cdot 0.20}{0.021} = \frac{0.010}{0.021} \approx 0.4762 \text{ quad } (47.62\%)}$$

Notice the dramatic Bayesian update: Factory 3 only produces 20% of speakers, but accounts for nearly **48%** of all defective units!

4. Python Implementation

```
import numpy as np

p_prior = np.array([0.50, 0.30, 0.20])
p_likelihood = np.array([0.01, 0.02, 0.05])

# Marginal evidence
p_evidence = np.sum(p_prior * p_likelihood)

# Posterior distribution
p_posterior = (p_prior * p_likelihood) / p_evidence

print(f"Total Evidence P(D): {p_evidence:.4f}")
print(f"Prior Probabilities: {p_prior}")
print(f"Posterior Probabilities: {np.round(p_posterior, 4)}")
print(f"P(Factory 3 | Defect): {p_posterior[2]:.4f} (47.62%)")
```

5. Key Takeaways & Interview Points

- **Belief Updating:** $\text{Posterior} \propto \text{Prior} \times \text{Likelihood}$.
- **Naive Bayes Classifier:** Foundation of text classification, assuming features are conditionally independent given class labels.